Q1. Consider, $\frac{dx}{dt} = -x^3 + 5x^2 - 6x = f(x)$.

for equilibrium solution put

$$\frac{dx}{dt} = 0 \Rightarrow -x^3 + 5x^2 - 6x = 0 \Rightarrow -x(x^2 - 5x + 6) = 0 \Rightarrow x = 0, 2, 3.$$

For, $x = 0$, $f'(x) = -3x^2 + 10x - 6, f'(0) = -6 < 0$.

$x = 0$ is the **stable equilibrium point**.

For, $x = 2$, $f'(x) = -3x^2 + 10x - 6, f'(2) = 2 > 0$.

$x = 2$ is the **unstable equilibrium point.**

For, $x = 3$, $f'(x) = -3x^2 + 10x - 6, f'(3) = -3 < 0$.

$x = 3$ is the **stable equilibrium point.**

2, Soln) $\frac{dy}{dx} = xy^{1/3}$ $\quad y(0) = 0$

To check whether $f(x,y) = xy^{1/3}$ is Lipschitz w.r.t "y" in any nbd of $y = 0$

$$\frac{|f(x,y_1) - f(x,y_2)|}{|y_1 - y_2|} = \frac{|xy_1^{1/3} - xy_2^{1/3}|}{|y_1 - y_2|} = \frac{|x||y_1^{1/3} - y_2^{1/3}|}{|y_1 - y_2|}$$

take $y_1 > 0$ and $y_2 = 0$.

$$= \frac{|x||y_1^{1/3}|}{|y_1|} = |x| \left|\frac{1}{y_1^{2/3}}\right| \to \infty \text{ as } y_1 \to 0$$

Thus Lipschitz condition is violated at any region that includes $y = 0$
also note that

$$\frac{\partial f}{\partial y} = \frac{1}{3} x y^{-2/3} \to \infty \quad \text{as } y \to 0$$

on solving $\frac{dy}{dx} = xy^{1/3}$, $y(0) = 0$

$$\Rightarrow y^{-1/3} dy = x dx \quad [\text{Variable Separable}]$$

$$\Rightarrow \frac{3}{2} y^{2/3} = \frac{x^2}{2} + C$$

$y(0) = 0$

$$\Rightarrow 0 = 0 + c \qquad \Rightarrow C = 0$$

$$\Rightarrow \frac{3}{2} y^{2/3}(x) = \left(\frac{x^2}{2}\right)$$

$$\Rightarrow y^{2/3} = \frac{x^2}{3}$$

$$\Rightarrow y = \left(\frac{x^2}{3}\right)^{3/2}$$ is a soln of I.V.P

also note that

$y(x) \equiv 0$ is also a soln of this differential eq$\underline{^n}$ which satisfy the initial condition.

Hence, options (a) & (d) are correct.

Problem 3. Let P be the total population.

Let $I(t)$ be the number of infected people at time t.

Let $N(t)$ " " " " non-infected " " " ".

Then $N(t) = P - I(t)$

Proportion of infected people, $y(t) = \dfrac{I(t)}{P}$

" " non-infected " $= \dfrac{P - I(t)}{P} = 1 - y(t)$

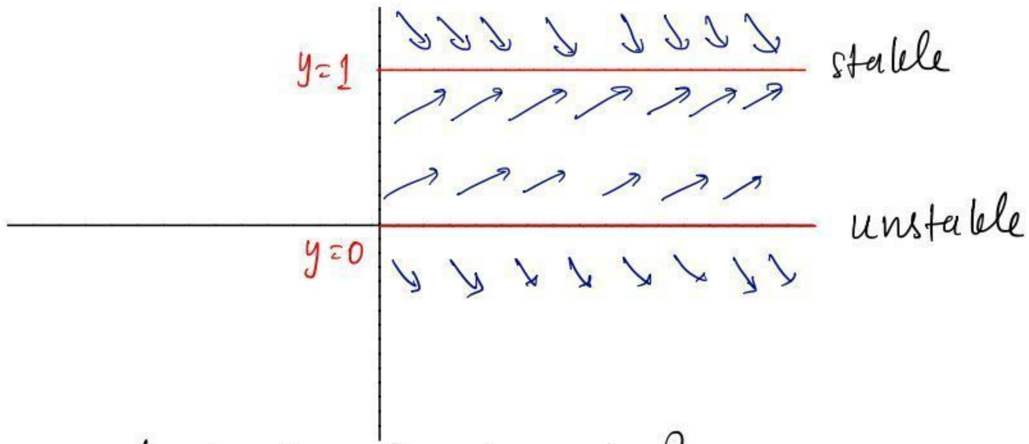
Given that $\dfrac{dI}{dt} \propto I(t)\, N(t)$

$$\Rightarrow \frac{dI}{dt} = k_1 I(t)\,(P - I(t))$$

$$\Rightarrow \cancel{P}\frac{dy}{dt} = \underbrace{k_1 P^{\cancel{2}}}_{k}\, y(t)\,(1 - y(t))$$

$$\Rightarrow \frac{dy}{dt} = k\, y(t)\,(1 - y(t)), \quad k > 0 \rightarrow \text{model}$$

Equilibrium solutions are $y \equiv 0$ & $y \equiv 1$.



$$y' = ky(1-y) = ky - ky^2$$

$$\Rightarrow y' - ky = -ky^2$$

Bernoulli with $p(t) = -k$, $g(t) = -k$, $a = 2$

$$u = y^{1-a} = y^{-1} \Rightarrow u' = -\frac{1}{y^2} y'$$

$$\Rightarrow u' = -\frac{1}{y^2}\left(ky - ky^2\right) = -\frac{k}{y} + k$$

$$\Rightarrow u' = -ku + k \Rightarrow u' + ku = k$$

So, $u = e^{-kt}\left(\int k e^{kt}\, dt + C\right)$

$$= e^{-kt}\left(e^{kt} + C\right) = 1 + Ce^{-kt}$$

$$\Rightarrow \boxed{y(t) = \frac{1}{u(t)} = \frac{1}{1 + Ce^{-kt}}}$$

$\lim_{t\to\infty} y(t) = 1 \Rightarrow$ In a long time, the whole population would be infected.

Problem 4.

$$x^2 + (y-C)^2 = 1 + C^2$$

$$\Rightarrow 2x + 2(y-C)y' = 0 \Rightarrow y' = -\frac{x}{y-C}$$

$$x^2 + y^2 + \cancel{C^2} - 2Cy = 1 + \cancel{C^2}$$

$$\Rightarrow C = \frac{x^2 + y^2 - 1}{2y} \Rightarrow y - C = y - \frac{x^2 + y^2 - 1}{2y} = \frac{y^2 - x^2 + 1}{2y}$$

$$\Rightarrow y' = -\frac{2xy}{y^2 - x^2 + 1} = f(x, y)$$

So, for orthogonal family, $y' = -\frac{1}{f(x,y)} = \frac{y^2 - x^2 + 1}{2xy}$

$$y' = \frac{1}{2x}y + \frac{1-x^2}{2x}\frac{1}{y} \Rightarrow y' - \frac{1}{2x}y = \frac{1-x^2}{2x}\frac{1}{y}$$

Bernoulli with $p(x) = -\frac{1}{2x}$, $g(x) = \frac{1-x^2}{2x}$, $a = -1$

$$u = y^{1-a} = y^2 \Rightarrow u' = 2yy' = 2y\left(\frac{1}{2x}y + \frac{1-x^2}{2x}\cdot\frac{1}{y}\right)$$

$$\Rightarrow u' = \frac{1}{x}y^2 + \frac{1-x^2}{x} \Rightarrow u' - \frac{1}{x}u = \frac{1-x^2}{x}$$

$$\Rightarrow u(x) = e^{-\int -\frac{1}{x}dx}\left(\int \frac{1-x^2}{x} e^{\int -\frac{1}{x}dx} dx + k\right)$$

$$= x\left(\int \frac{1-x^2}{x^2}dx + C\right) = x\left(-\frac{1}{x} - x + k\right)$$

$$\Rightarrow u(x) = -1 - x^2 + kx = y^2$$

$$\Rightarrow x^2 - kx + y^2 = -1 \Rightarrow \left(x - \frac{k}{2}\right)^2 + y^2 = \frac{k^2}{4} - 1$$

$$\Rightarrow \boxed{(x-k)^2 + y^2 = k^2 - 1}$$ (renaming $\frac{k}{2}$ as k)

Q - 5 $\quad y' + x^2 y = \dfrac{e^{-x^3}\sinh x}{3y^2}$

Soln) $\quad 3y^2 y' + 3x^2 y^3 = e^{-x^3}\sinh x$

Let $u = y^3 \qquad \dfrac{du}{dx} = u' = 3y^2 y'$

$\Rightarrow u' + 3x^2 u = e^{-x^3}\sinh x \rightarrow$ Linear D.E

$I.f = e^{3\int x^2 dx} = e^{x^3}$

$\Rightarrow u e^{x^3} = \int e^{x^3} \cdot e^{-x^3} \sinh x \, dx$

$\Rightarrow u e^{x^3} = \cosh x + c$

$\Rightarrow u = e^{-x^3}[\cosh x + c]$

put $u = y^3$

$\Rightarrow y^3 = e^{-x^3}[\cosh x + c]$ is the soln of given D.E where c is arbitrary constant.